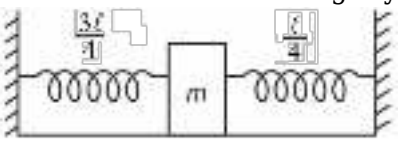
PHYSICS

1) **Statement-1 :** A particle is moving along x-axis. The resultant force F acting on it is given by F = –ax – b, where a and b are both positive constants. The motion of this particle is not SHM.
Statement-2 : In SHM magnitude of resultant force must be proportional to the displacement from mean position.

(1) Statement-1 is True, Statement-2 is True.
(2) Statement-1 is False, Statement-2 is False.
(3) Statement-1 is True, Statement-2 is False
(4) Statement-1 is False, Statement-2 is True

2) A spring of stiffness constant k and natural length is cut into two parts of $\frac{3\ell}{4}$ $\frac{\ell}{4}$ lengths and respectively and an arrangement is made as shown. If the mass is slightly displaced and then released, what is the time period of oscillation?



(1) $2\pi\sqrt{\frac{3m}{2k}}$
(2) $2\pi\sqrt{\frac{3m}{16k}}$
(3) $2\pi\sqrt{\frac{3m}{4k}}$
(4) $\frac{\pi}{2}\sqrt{\frac{m}{3k}}$



3) Find time taken by the particle to go from position 1 to position 2.

(1) $\frac{7T}{16}$
(2) $\frac{7T}{12}$
(3) $\frac{7T}{6}$
(4) $\frac{7T}{24}$

4) If equation of SHM are given as $x_1 = 4\sin\omega t$ and $x_2 = 4\cos\omega t$ and both SHM are superimposed then we will get resultant SHM. Find total energy in resultant SHM (If K = 100 N/m for both the equation)

(1) 1600 J

(2) 800 J

(3) 400 J

(4) 200 J

5) A particle performs simple harmonic motion with amplitude A. Its speed is tripled at the instant when it is at distance $\frac{2A}{3}$ from equilibrium position. The new amplitude of the motion is:

(1) 3A

(2) $A\sqrt{3}$

(3) $\frac{7A}{3}$

(4) $\frac{A}{3}\sqrt{41}$

6) For a sine wave passing through a medium, let y be the displacement of a particle, v be its velocity and a be its acceleration :-

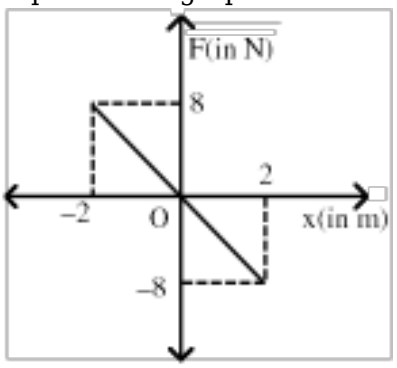
(1) y and a are always in opposite phase

(2) Phase difference between y and v is π/2

(3) Phase difference between v and a is π/2

(4) All of the above

7) A particle of mass 0.01 kg is executing SHM about x = 0 under the influence of a force. The force-displacement graph is shown in diagram. The time period of the SHM is :-



(1) 1.05 sec

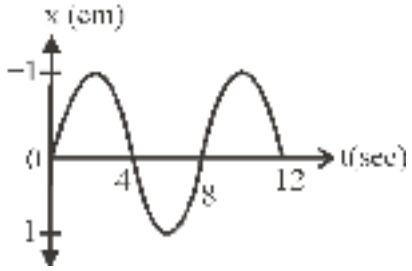
(2) 0.5 sec

(3) 0.25 sec

(4) 0.3 sec

8) The x-t graph of a particle undergoing simple harmonic motion is shown in figure. The

acceleration of the particle at t = $\frac{4}{3}$s is :-



(1) $\frac{\sqrt{3}}{32}\pi^2 cm/s^2$

(2) $\frac{-\pi^2}{32} cm/s^2$

(3) $\frac{\pi^2}{32} cm/s^2$

(4) $-\frac{\sqrt{3}}{32}\pi^2\, cm/s^2$

9) A simple pendulum is executing simple harmonic motion with a time period T. If the length of the pendulum is increased by 21%, the increase in the time period of the pendulum of increased length is:-

(1) 10%

(2) 21%

(3) 30%

(4) 50%

10) Two tuning forks of frequencies n_1 and n_2 produce n beats per second. If n_2 and n are known, n_1 may be given by :-

(1) $\frac{n_2}{n} + n_2$

(2) $n_2\, n$

(3) $n_2 \pm n$

(4) $\frac{n_2}{n} - n_2$

11) The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is :-

(1) 13.3 cm

(2) 8 cm

(3) 12.5 cm

(4) 16 cm

12) Waves from two sources superimpose on each other at a particular point. Amplitude and frequency of both the waves are equal. The ratio of intensities when both waves reach in the same phase and they reach with the phase difference of 90° will be

(1) 1:1

(2) √2:1

(3) 2:1

(4) 4:1

13) The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is :

(1) 44%

(2) 40%

(3) 20%

(4) 10%

14) In stationary waves, antinodes are the points where there is -

(1) Minimum displacement and minimum pressure change

(2) Minimum displacement and maximum pressure change

(3) Maximum displacement and maximum pressure change

(4) Maximum displacement and minimum pressure change

15) The equation of a simple harmonic wave is given by :-

$y = 3 \sin \frac{\pi}{2}(50\ t - x)$,

where x and y are in metres and t is in seconds. The ratio of maximum particle velocity to the wave velocity is :-

(1) 3π

(2) $\frac{2}{3\pi}$

(3) 2π

(4) $\frac{3}{2\pi}$

16) **Statement-1:** Two waves given by equations : $y_1(x, t) = 2a \sin(\omega t - kx)$ and $y_2(x, t) = a \sin(2\omega t - 2kx)$ will have equal intensity.

Statement-2: Intensity of waves propagating in same medium is proportional to square of amplitude only.

(1) Statement-1 is false, statement-2 is true

(2) Statement-1 is true, statement-2 is false

(3) Statement-1 is false, statement-2 is false

(4) Statement-1 is true, statement-2 is true

17) A second harmonic has to be generated in a string of length L stretched between two rigid supports. The point where the string has to be plucked and touched are :-

(1) plucked at $\frac{L}{4}$ and touch at $\frac{L}{2}$

(2) plucked at $\frac{L}{4}$ and touch at $\frac{3L}{2}$

(3) plucked at $\frac{L}{2}$ and touch at $\frac{L}{2}$

(4) plucked at $\frac{L}{2}$ and touch at $\frac{3L}{2}$

18) A rope of length L and mass M hangs freely from the ceiling. If time taken by a transverse wave to travel from the bottom to the top of the rope is T, then time to cover first half length is :

(1) T

(2) $T\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$

(3) $\frac{T}{\sqrt{2}}$

(4) T/2

19) When two tuning forks (fork 1 and fork 2) are sounded simultaneously, 4 beats per second are heard. Now, some wax is attached on the prong of fork 2. When the tuning forks are sounded again, 6 beats per second are heard. If the frequency of fork 1 is 200 Hz, then what was the original frequency of fork 2 ?

(1) 196 Hz

(2) 204 Hz

(3) 200 Hz

(4) 202 Hz

20) Equation of travelling wave is given by
$y = 10 \sin\left[2\pi t - \frac{\pi}{2}x + \frac{\pi}{6}\right]$ m.
Find out time difference between particles placed at x = 2m and x = 5m.

(1) $\frac{4}{3}$ sec.

(2) $\frac{3}{4}$ sec.

(3) $\frac{1}{4}$ sec.

(4) $\frac{1}{3}$ sec.

21) Which of the following equations can represent an equation of progressive wave :
(a) y = A tan(ωt – kx)
(b) y = A cos(ωt – kx)

(c) $y = \frac{1}{t + x}$

(d) $y = e^{-(x-vt)^2}$

(e) $y = Ae^{x} \cos(\omega t - \theta)$

(1) a, c

(2) b, d, e

(3) b, d

(4) d, e

22) Amplitude of a damped oscillator becomes half every 2mins. Amplitude after 8 minutes be 1/x of the original. Find x.

(1) $\frac{1}{8}$

(2) 8

(3) 3

(4) $\frac{1}{3}$

23) Two liquids are at temperature 20°C and 40°C. When same mass of both of them is mixed then temperature of mixture is 32°C. What is ratio of their specific heat ?

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{1}{5}$

(4) $\frac{2}{5}$

24) A student is performing the experiment of resonance column. The diameter of the column tube is 6 cm. The frequency of the tunning fork is 504 Hz speed of the sound at the given temperature is 336 m/s. The zero of the meter scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is :-

(1) 13 cm

(2) 16.6 cm

(3) 18.4 cm

(4) 14.8 cm

25) A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer is in contact with the object reads $x_0/2$?

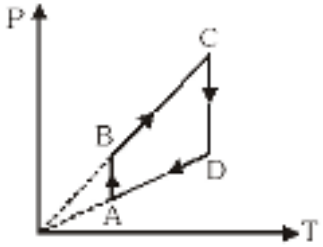
(1) 35

(2) 25

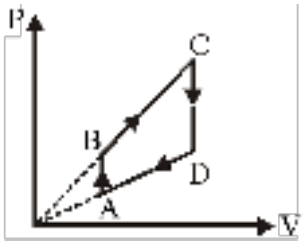
(3) 60

(4) 40

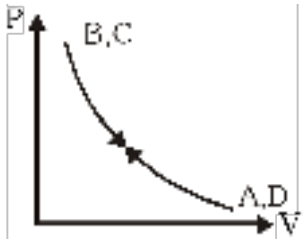
26) Pressure versus temperature graph of an ideal gas is as shown in figure. Corresponding pressure (P) versus volume (V) graph will be-



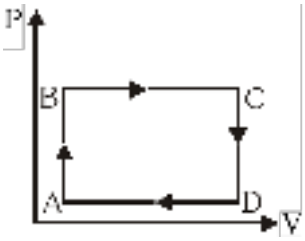
(1)



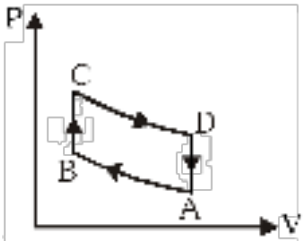
(2)



(3)



(4)



27) A solid sphere and a hollow sphere of the same material and of equal radii are initially at the same temperature and both are kept in common surrounding which is at lower temperature.

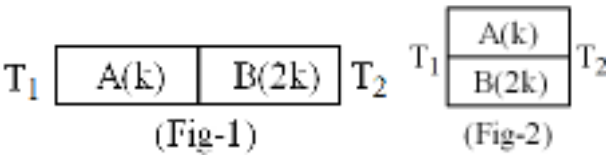
(1) Hollow spheres will emit thermal radiation at faster rate in the beginning

(2) Both the spheres will emit thermal radiation at equal rates at all instants of time

(3) The rate of cooling dT/dt will be the same for the two spheres

(4) Rate of cooling is faster for hollow sphere

28) Two identical rods A and B of different materials are welded together as shown in figure 1 & 2. Their thermal conductivities are k and 2k. The ratio of equivalent thermal conductivity of the composite rods, when connected in series and in parallel connection, is



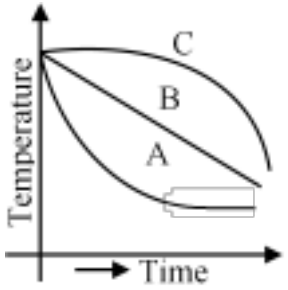
(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{8}{9}$

(4) $\frac{9}{4}$

29) A block of steel heated to 100°C is left in a room to cool. Which of the curves shown in fig. represents the correct behavior ?



(1) A

(2) B

(3) C

(4) None of these

30) A liquid of mass 2m and specific heat $\frac{C}{2}$ is heated to a temperature T. Another liquid of mass $\frac{m}{3}$ and specific heat 2C is heated to a temperature 3T. If these two liquids are mixed, the resulting temperature of the mixture is:

(1) $\frac{3T}{5}$

(2) $\frac{9T}{5}$

(3) $\frac{5T}{3}$

(4) $\frac{2T}{3}$

31) A vessel has 6 g of hydrogen at pressure P and temperature 500K. A small hole is made in it so that hydrogen leaks out. How much hydrogen leaks out if the final pressure is P/2 and temperature falls to 300K?

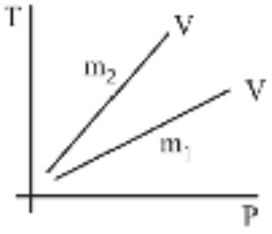
(1) 2g

(2) 3g

(3) 4g

(4) 1g

32) Below two graphs are drawn for an ideal gas at equal volume, then mass of gas samples m_1 & m_2 are related as:



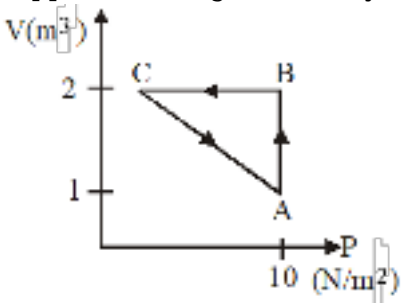
(1) $m_1 > m_2$

(2) $m_1 < m_2$

(3) $m_1 = m_2$

(4) None

33) An ideal gas is taken through the cycle A→B→C→A, as shown in the figure. If the net heat supplied to the gas in the cycle is 5 J, the work done by the gas in the process C→A is :-



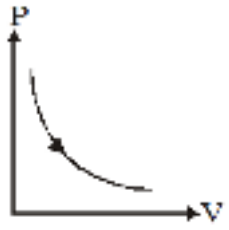
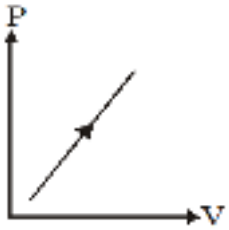
(1) –5 J

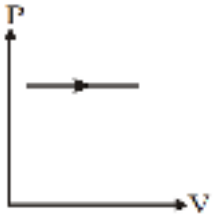
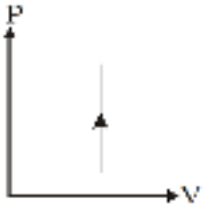
(2) –10 J

(3) –15 J

(4) –20 J

34) Match correctly :-

(a)		(p)	Volume is constant
(b)		(q)	Volume is increasing

(c)		(r)	Temperature is constant
(d)		(s)	Pressure is constant

(1) a→p ; b→q,r ; c→q ; d→p

(2) a→r, p ; b→q,r ; c→q,s ; d→p,r

(3) a→q, r ; b→q ; c→q,s ; d→p

(4) None

35) A copper block of mass 4 kg is heated in a furnace to a temperature 425°C and then placed on a large ice block. The mass of ice that will melt in this process will be (Specific heat of copper = 500 J kg^{-1} $°C^{-1}$ and heat of fusion of ice = 336 kJ kg^{-1}) :-

(1) 0.5 kg

(2) 1 kg

(3) 1.5 kg

(4) 2.5 kg

36) One mole of ideal monoatomic gas ($\gamma = 5/3$) is mixed with one mole of diatomic gas ($\gamma = 7/5$). What is γ for the mixture if γ denotes the ratio of specific heat at constant pressure, to that at constant volume?

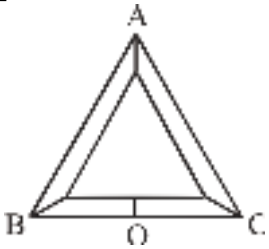
(1) 3/2

(2) 23/15

(3) 35/23

(4) 4/3

37) Three rods of equal length are joined to form an equilateral triangle ABC. O is the mid point of BC. Distance OA remains same for small change in temperature. Coefficient of linear expansion for AB and AC is same, i.e., α_2 but that for BC is α_1. Then :-



(1) $\alpha_2 = \alpha_1 / 2$

(2) $\alpha_2 = 2\alpha_1$

(3) $\alpha_2 = \frac{\alpha_1}{3}$

(4) $\alpha_2 = \frac{\alpha_1}{4}$

38) Solar radiation emitted by the sun resembles that emitted by a black body at a temperature of 6000 K. Maximum intensity is emitted at wavelength of about 4800 Å. If the sun was cooled down from 6000 K to 3000 K, then the peak intensity would occur at a wavelength of :-

(1) 4800 Å

(2) 9600 Å

(3) 2400 Å

(4) 19200 Å

39) The temperature of n mol of an ideal gas is increased from T to 4T though a process $P = aT^{-1}$ when a is a constant. Then work done by the gas is :-

(1) nRT

(2) 4nRT

(3) 2nRT

(4) 6nRT

40) Three very large plates of same area are kept parallel and close to each other. They are considered as ideal black surfaces and have very high thermal conductivity. The first and third plates are maintained at temperature 2T and 3T respectively. The temperature of the middle (i.e. second) plate under steady state condition is :-

(1) $\left(\frac{65}{2}\right)^{1/4}$ T

(2) $\left(\frac{97}{4}\right)^{1/4}$ T

(3) $\left(\frac{97}{2}\right)^{1/4}$ T

(4) $(97)^{1/4}$ T

41) 1 mole of an ideal gas at 300 K is subjected to a reversible adiabatic expansion to double its volume. The change in internal energy ($\gamma = 1.5$)

(1) 1169 J

(2) 769 J

(3) 1462 J

(4) 969 J

42) A body cools from 70ºC to 50ºC in 10 min. The surrounding temperature is 30ºC. In what further time (in minutes) will it cool to 40ºC?

(1) 10 min

(2) 5 min

(3) 15 min

(4) infinite

43) Two particles execute SHM of the same amplitude and frequency along the same straight line. If they pass one another when going in opposite directions, each time their displacement is half their amplitude; then the phase difference between them is :-

(1) $\frac{\pi}{3}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{6}$

(4) $\frac{2\pi}{3}$

44) Force constant of a spring is K. If 1/5th part is detached then force constant of remaining spring will be :

(1) $\frac{4}{5}K$

(2) $\frac{5}{4}K$

(3) $\frac{K}{5}$

(4) 5K

45) Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperatures 200°C and 20°C, respectively. the temperature of junction B will be :-



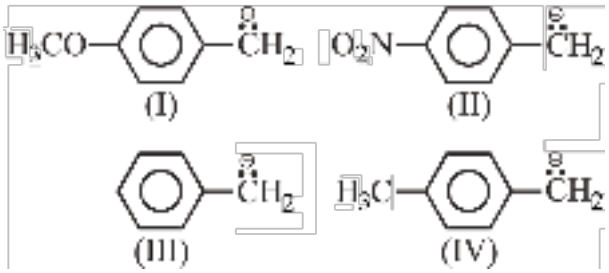
(1) 120°C

(2) 100°C

(3) 140°C

(4) 80°C

CHEMISTRY

1) Consider the following carbanions:-  Correct decreasing order of stability is :

(1) III > IV > I > II

(2) II > III > IV > I

(3) IV > I > II > III

(4) I > II > III > IV

2) Total possible isomers of C_4H_8 are

(1) 4

(2) 5

(3) 6

(4) 7

3)

The reagent needed to convert 2-butyne to cis-2-butene is :-

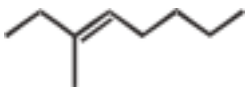
(1) H_2/Pt

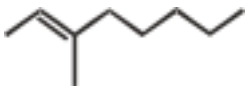
(2) H_2/Lindlar's catalyst

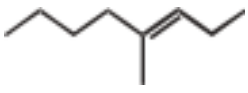
(3) Li/NH_3

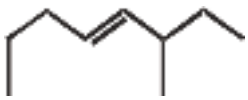
(4) None of these

4) Alkene (X)  X is :-

(1) 

(2) 

(3) 

(4) 

5)

CH_3–CH=CH_2 $\xrightarrow{Br_2/NaCl}$

Product of the reaction will be :-

(1) Only 1, 2-dibromopropane

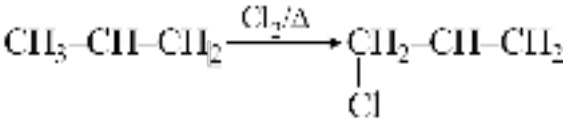
(2) Only 1-bromo-2-chloropropane

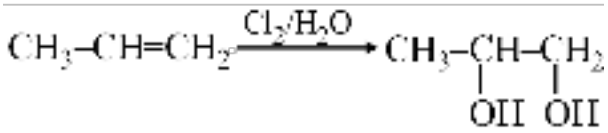
(3) Only-2-bromo-1-chloropropane

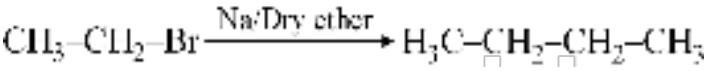
(4) Mixture of 1, 2 dibromopropane and 1-bromo-2-chloropropane

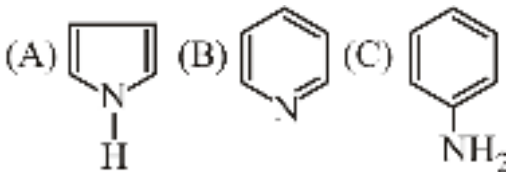
6) Which of the following reactions is incorrect ?

(1) $CH_3–CH_2–COONa \xrightarrow{\text{Electrolysis}} CH_3–CH_2–CH_2–CH_3$

(2) 

(3) 

(4) 

7) 

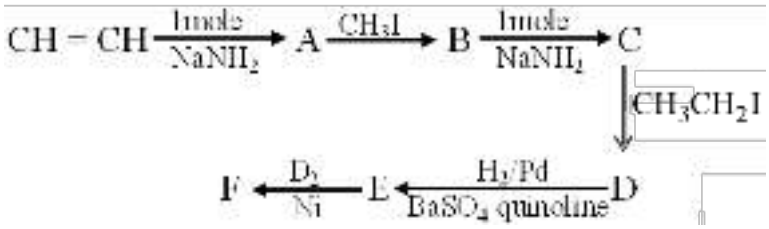
Choose the incorrect statement -

(1) A is more basic than B

(2) C is more basic than A

(3) B is more basic than C

(4) C have delocalised electron pair at 'N'

8) 

Incorrect statement about the conversions given :-

(1) D to E conversion is a syn-addition of H_2

(2) F is a meso compound and is optically inactive

(3) B can gives red precipitate with ammonical cuprous chloride

(4) C to D conversion is a nucleophilic substitution and follows SN^2 mechanism

9)

The IUPAC name of the compound $CH_3CH=CHC\equiv CH$ is :

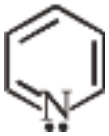
(1) Pent-3-en-1-yne

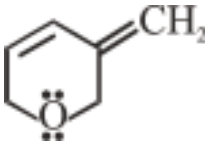
(2) Pent-2-en-4-yne

(3) Pent-1-yn-3-ene

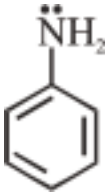
(4) Pent-4-yn-2-ene

10) In which of the following cases, lone-pair is involved in resonance :-

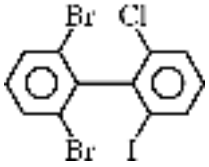
(1) 

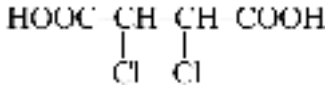
(2) 

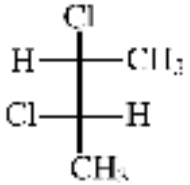
(3) 

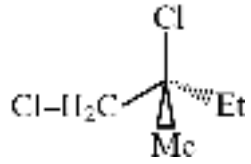
(4) 

11) Which of the following statement is correct ?

(1) 
is optically active.

(2) 
shows optical isomerism.

(3) 
is optically active.

(4) 
has S-configuration

12)

Which of the following statement is false ?

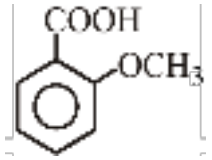
(1) cis-2-butene and trans-2-butene are diastereomers.

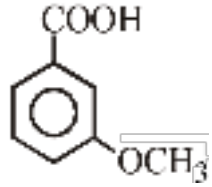
(2) Dihedral angle in staggered form of ethane is 60°

(3) Conformational isomers are non resolvable at room temperature

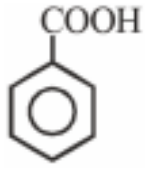
(4) Energy of anti form of n-butane is more than its gauche form

13) Arrange the following in order of acidic strength?

(P) 

(Q) 

(R) 

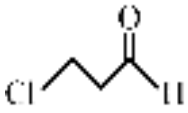
(S) 

(1) P > R > Q > S

(2) Q > S > P > R

(3) P > Q > S > R

(4) R > Q > P > S

14) IUPAC name of  is :

(1) 4–chloro butan–2–one

(2) 3–chloro propanal

(3) 1–chloro propan–3–al

(4) 1–chloro butan–3–al

15) The geometry of the compound is :-

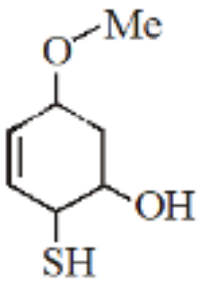
$H_2C = CH - CH = CH_2$

(1) Tetrahedral

(2) Linear

(3) Planar

(4) Pyramidal

16) The IUPAC name of given compound is :- 

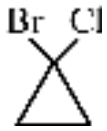
(1) 3-methoxy-6-mercaptocyclohex-4-enol

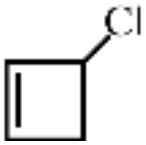
(2) 6-hydroxy-4-methoxycyclohex-2-ene-1-thiol

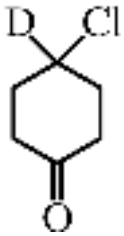
(3) 5-methoxy-2-mercaptocyclohex-3-en-1-ol

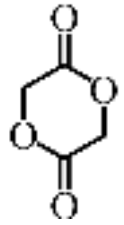
(4) 2-mercapto-5-methoxycyclohex-3-en-l-ol

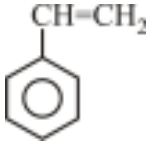
17) Which of the following compound is chiral ?

(1) 

(2) 

(3) 

(4) 

18) The ratio of σ and π bonds in styrene 

is :-

(1) 4 : 1

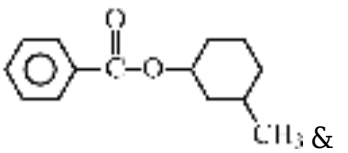
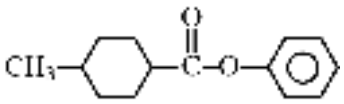
(2) 1 : 4

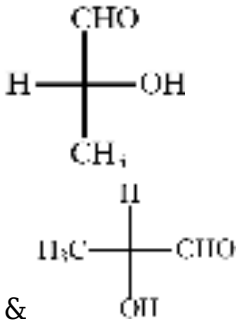
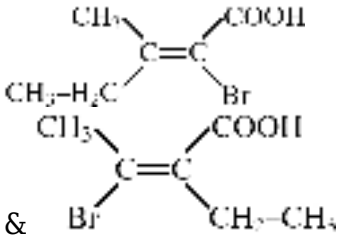
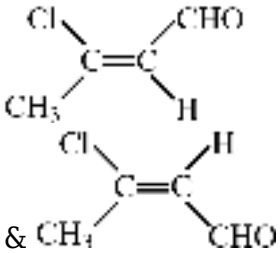
(3) 5 : 2

(4) 10 : 1

19)

Match the column

Column-I (Compounds)		Column-II (Isomerism)	
(a)	 & 	(P)	Enantiomers

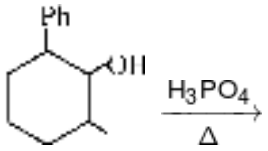
(b)	 &	(Q)	Metamers
(c)	 &	(R)	Diastereomers
(d)	 &	(S)	Chain isomers

(1) a→S, b→R, c→Q, d→P

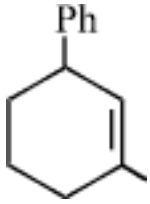
(2) a→Q, b→P, c→R, d→S

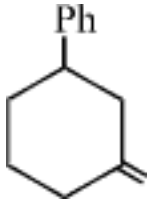
(3) a→Q, b→R, c→P, d→S

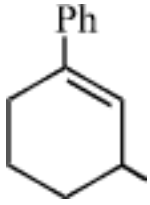
(4) a→Q, b→P, c→S, d→R

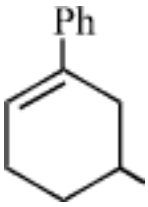
20) 

Major product obtained by dehydration of given alcohol is :-

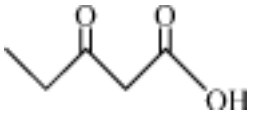
(1) 

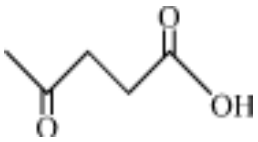
(2) 

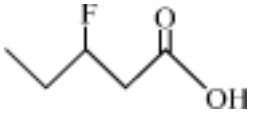
(3) 

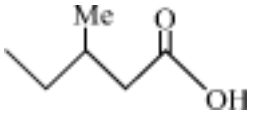
(4) 

21) Most reactive for decarboxylation is :-

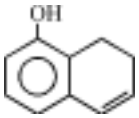
(1) 

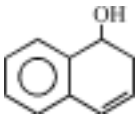
(2) 

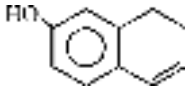
(3) 

(4) 

22) Most reactive alcohol toward acid catalysed dehydration is:-

(1) 

(2) 

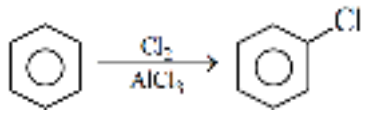
(3) 

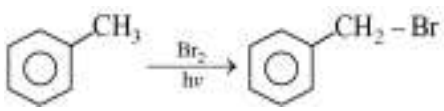
(4) 

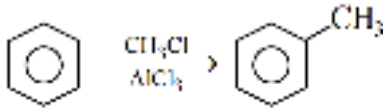
23) The compound with Highest-Boiling point is ?

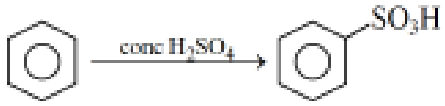
(1) n-Heptane
(2) n-Pentane
(3) 2, 2-Dimethyl propane
(4) n-butane

24) Which of the following is not example of electrophilic substitution reaction ?

(1) 

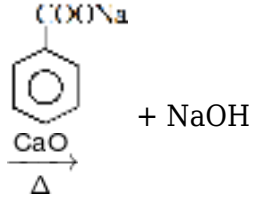
(2) 

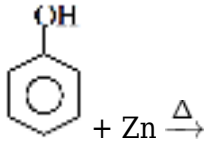
(3) 

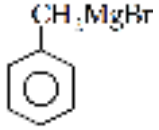
(4) 

25) Which of the following reaction doesn't produce benzene ?

(1) $HC \equiv CH \xrightarrow[\text{Fe Tube}]{\text{Red Hot}}$

(2) 

(3) 

(4) 

26) Which of the following is chain termination step ?

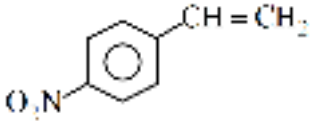
(1) $Cl \cdot Cl \xrightarrow{h\nu} \dot{Cl} + \dot{Cl}$

(2) $CH_3Cl + \dot{Cl} \longrightarrow \dot{C}H_2Cl + HCl$

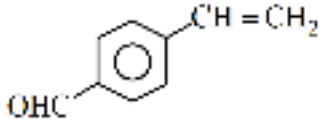
(3) $\dot{C}H_3 + Cl_2 \longrightarrow CH_3Cl + \dot{Cl}$

(4) $\dot{Cl} + \dot{Cl} \longrightarrow Cl_2$

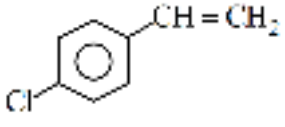
27) Rate of electrophilic addition reaction is ?



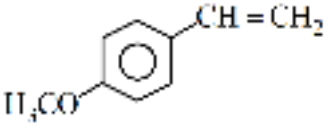
(i)



(ii)



(iii)



(iv)

(1) iv > iii > i > ii

(2) iv > iii > ii > i

(3) i > ii > iii > iv

(4) i > ii > iv > iii

28) Order of bond dissociation energy of C – H bond is.
(i) CH_3 – H
(ii) CH_3 – CH_2 – H
(iii) CH_2 = CH – CH_2 – H
(iv) H_2C = CH – H

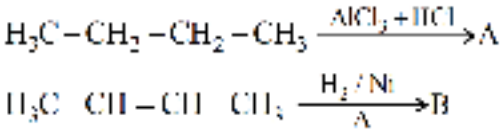
(1) iii > i > ii > iv
(2) iii > ii > i > iv
(3) iv > ii > i > iii
(4) iv > i > ii > iii

29) Which of the following doesn't give test of unsaturation ?

(1) 
(2) 
(3) HC ≡ CH
(4) CH_2 = CH_2

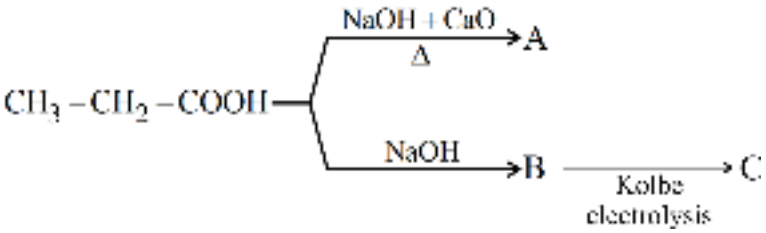
30) Which of the following give red precipitate on reaction with ammonical cuprous chloride ?

(1) C_4H_{10}
(2) C_5H_{10}
(3) C_3H_4
(4) C_2H_6

31) 

A and B are ?

(1) Position isomer
(2) Chain isomer
(3) Functional group isomer
(4) Homologues

32) 

Which is incorrect ?

(1) A is ethane

(2) B is CH_3CH_2COONa

(3) C is propane

(4) C is butane

33) Which of the following reaction occur at slowest rate ?

(1) $CH_4 + F_2 \xrightarrow{dark}$

(2) $CH_4 + Cl_2 \xrightarrow{h\nu}$

(3) $CH_4 + Br_2 \xrightarrow{h\nu}$

(4) $CH_4 + I_2 \xrightarrow{h\nu}$

34) Which of the following compound is used as an adsorbent in column chromatography :-

(1) Na_2O

(2) Na_2SO_4

(3) Al_2O_3

(4) NaCl

35) Which of the following compound can be separated by steam distillation method.:-

(1) Steam volatile but insoluble in water

(2) Steam volatile but soluble in water

(3) Steam non volatile but sparingly soluble in water

(4) Liquid in steam but solid in water

36) Lassaigne's test is used for the detection of :-

(1) N, S and halogens

(2) C, H and P

(3) C, H and O

(4) C, S and P

37) Which pair of resonating structures is correct :-

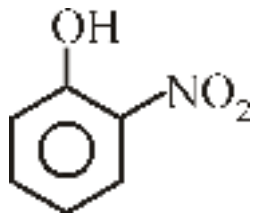
(1) $H_2\overset{\ominus}{C}-\overset{\cdot\cdot}{N}H_2 \leftrightarrow H_2C=\overset{\oplus}{N}H_2$

(2) $H_2C=CH-CH=O \leftrightarrow H_2\overset{\ominus}{C}-CH-CH=\overset{\oplus}{O}$

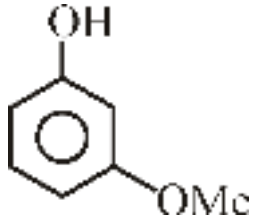
(3) $H_2C=CH-\dot{C}H_2 \leftrightarrow H_2\dot{C}\ CH=CH_2$

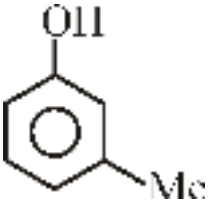
(4) All of these

38) The decreasing order of acidic strength is :-

(a) 

(b) 

(c) 

(d) 

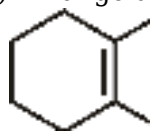
(1) a > b > c > d

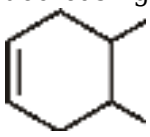
(2) d > c > b > a

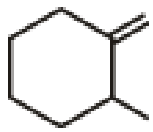
(3) b > a > c > d

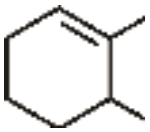
(4) c > a > b > d

39) Arrange the following in decreasing order of stability of alkene.

(a) 

(b) 

(c) 

(d) 

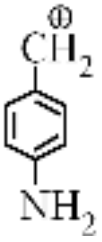
(1) a > b > c > d

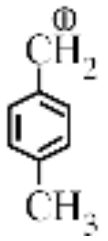
(2) d > c > b > a

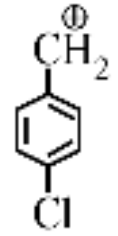
(3) a > b > d > c

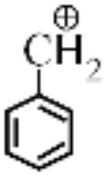
(4) a > d > b > c

40) Which is most stable carbocation ?

(1) 

(2) 

(3) 

(4) 

41) Which species is not a nucleophile?

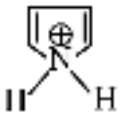
(1) 

(2) R–O–H

(3) 

(4) 

42) Match the column

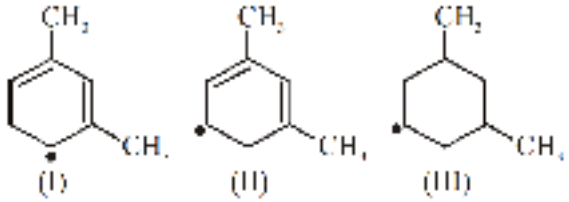
Compound		Type	
(A)		(P)	Aromatic
(B)		(Q)	Non aromatic
(C)		(R)	Antiaromatic

(1) A→Q, B→P, C→Q

(2) A→P, B→Q, C→R

(3) A→R, B→Q, C→P

(4) A→R, B→P, C→Q

43) 

Correct order of stability of these radicals ?

(1) I > II > III

(2) III > II > I

(3) II > I > III

(4) I > III > II

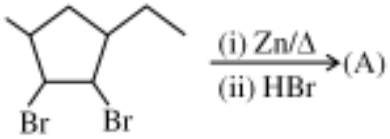
44) **Statement-I :-** Acetate ion have equivalent bond of both "C–O" bonds.
Statement-II :- Acetic acid shows resonance.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

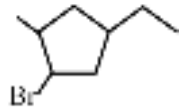
45) 

Major product

Identify product A

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) First action spectrum of photosynthesis was described after experiments with the help of :-

(1) *Chlamydomonas*

(2) *Chlorella*

(3) DCPIP

(4) *Cladophora*

2) Structure formed by arrangement of stacks of flattened membranous sacs in chloroplast are -

(1) Fret channels

(2) Cristae

(3) Grana

(4) Mesosomes

3) Which type of chlorophyll are found in Higher plants ?

(1) Chl a, Chl c

(2) Chl a, Chl d

(3) Chl a, Chl b

(4) Chl a, Chl e

4) The graphic curve depicting the relative rates of photosynthesis at different wave lengths of light is called :

(1) Absorption spectrum

(2) Action spectrum

(3) Atomic spectrum

(4) Emission spectrum

5) Light reaction involves -

(1) Release of CO_2

(2) Formation of $NADH_2$

(3) Formation of ATP

(4) Rubisco enzyme

6) Which of the following hydrogen carrier pump H^+ into lumen

(1) PQ

(2) Pheophytin

(3) Plastocyanin

(4) FRS

7) During splitting of water, the protons and O_2 formed are likely to be released in/on –

(1) Outer side of the membrane

(2) Lumen

(3) Stroma

(4) Both Lumen and stroma

8) Match List I with List II and select the correct option.

List- I		**List- II**	
A.	NADP reductase enzyme	(i)	Transmembrane channel
B.	CF_0	(ii)	Inner side of the membrane
C.	CF_1	(iii)	Stroma side of membrane
D.	Splitting of the water molecule	(iv)	Protrudes on the outer surface of the thylakoid membrane

(1) A – ii, B – iv, C – iii, D – i

(2) A – iii, B – i, C – iv, D – ii

(3) A – iv, B – iii, C – ii, D – i

(4) A – iii, B – ii, C – iv, D – i

9) **Statement-I :** Calvin cycle occurs in all photosynthetic plants, it does not matter whether they have C_3 or C_4 pathways.
Statement-II : Carboxylation is the fixation of CO_2 into a stable organic intermediate.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct and Statement-II are incorrect.

(3) Statement-I is incorrect and Statement-II are correct.

(4) Both Statement-I and Statement-II are correct.

10) Fill in the blanks in the given statement.
"First CO_2 acceptor in C_4 plants is ...(i)... in .....(ii)..... cells, while second CO_2 acceptor is .....(iii) in .....(iv).... cells".

(1) (i) PEP of 3C (ii) bundle sheath (iii) RuBP of 5C (iv) Mesophyll

(2) (i) PEP of 3C (ii) Mesophyll (iii) RuBP of 5C (iv) bundle sheath

(3) (i) RuBP of 5C (ii) Mesophyll (iii) PEP of 3C (iv) bundle sheath

(4) (i) OAA of 4C (ii) Mesophyll (iii) PGA of 3C (iv) bundle sheath

11) **Assertion :-** Photorespiration is a wasteful process.
Reason :- In photorespiration, there is neither synthesis of sugar, nor ATP.

(1) If both assertion and reason are correct but reason is correct explanation of assertion

(2) If both assertion and reason are correct but reason is incorrect explanation of assertion

(3) Assertion and reason both are correct.

(4) Assertion and reason is incorrect.

12) Consider the following statements & choose the correct option.
(a) External factors affecting photosynthesis include availability of sunlight, water & chloroplast.
(b) When several factors affect any bio-chemical process. Blackman's law of limiting factor comes in to effect.
(c) In aquatic conditions generally water is available at suboptimal level.
(d) CO_2 is the main factor which limits the rate of photosynthesis on a clear day.
Options :-

(1) a, b, c, d are correct

(2) b, c, d are correct

(3) b and d are correct

(4) Only d is correct

13) Except for plants in shade, light is _____a limiting factor in nature. _____ in incident light beyond a point causes the _____of chlorophyll
Choose correct terms for blanks respectively :-

(1) Rarely, Increase, Break down

(2) Always, Increase, Break down

(3) Rarely, Decrease, Break down

(4) Always, Decrease, Break down

14) Identify the correct statements :
A. Light reaction or photochemical phase includes light absorption, water splitting and oxygen release only.
B. Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase.
C. Cyclic photophosphorylation also occurs when only light of wavelengths beyond 680 nm are available for excitation.
D. Peak of light absorption of the reaction center is same in both the photosystems.
Choose the **correct** answer from the options given below.

(1) A and D only

(2) A, B and D only

(3) B and C only

(4) A, B and C only

15) Identify the correct statements.
A. In C_3 plants first CO_2 acceptor was a five carbon ketose sugar.
B. Plants that can be adapted to dry tropical regions may have the C_4 pathway.
C. C_4 Plants shows kranz anatomy in their leaves.
D. In photorespiration RuBisCO shows carboxylase activity.
Choose the correct answer from the options given belove

(1) A and D only

(2) A, B and D only

(3) B and C only

(4) A, B and C only

16) The energy during cellular respiration is released in a series of slow step wise reactions controlled by :-

(1) Carbohydrate

(2) Fat

(3) Both (1) and (2)

(4) Enzymes

17) Where does glycolysis takes place in prokaryotic cells ?

(1) Nucleus

(2) Cytoplasm

(3) Mitochondria

(4) Ribosome

18) Which one of the following is not the cofactor of link reaction ?

(1) Mg^{2+}

(2) CoA

(3) NAD^{+}

(4) Ca^{+2}

19) How many ATP's are directly produced during single T.C.A cycle ?

(1) 1

(2) 2

(3) 4

(4) 6

20) Ubiquinone recieves reducing equivalents from :

(1) NADH

(2) $FADH_2$

(3) Both (1) & (2)

(4) ATP

21) How many ATP are produced through E.T.S during oxidation of one molecule of Glucose.

(1) 8

(2) 24

(3) 6

(4) 34

22) Fatty acid broken down to _____ before entering the respiratory pathway.

(1) Pyruvic acid

(2) Acetyl CoA

(3) PGAL

(4) α - Ketoglutaric acid

23) Find out the A, B, C and D in the following equation of lipid oxidation.
A $(C_{51}H_{98}O_6)$ + B $O_2 \rightarrow$ C CO_2 + D H_2O

(1) (A = 145), (B = 2), (C = 102), (D = 98)

(2) (A = 2), (B = 102), (C = 145), (D = 98)

(3) (A = 2), (B = 145), (C = 102), (D = 98)

(4) (A = 2), (B = 145), (C = 98), (D = 102)

24) Read the following statements carefully and select the correct option from given below.
Statement – I : In fermentation less than seven percent energy of the energy in glucose is released.
Statement – II : Fermentation process is hazardous, either alcohol or acid is produced.

(1) Both statements I and II are correct

(2) Statement I is correct but II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statements I and II are incorrect

25) **Assertion :** In plants, the glucose is derived from sucrose or starch.
Reason : Sucrose is converted into glucose and fructose by the enzyme amylase.
In the light of the above statements, choose correct answer from options given below :

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

26) Which of the following statement is incorrect ?

(1) Aerobic respiration takes place in the mitochondria.

(2) Link reaction catalysed by pyruvate decarboxylase enzyme.

(3) Pyruvate dehydrogenase require the participation of several co-enzymes.

(4) During link reaction, two molecules of NADH are produced from the metabolism of two molecules of pyruvic acid.

27) The movement of electrons from $FADH_2$ occurs via complex :-
(a) Complex II - Succinate dehydrogenase
(b) Complex I - NADH dehydrogenase
(c) Complex III - Cytochrome bc_1
(d) Complex IV - Cytochrome C oxidase

(1) a → c → d

(2) b → c → d

(3) a → b → c → d

(4) a → d → c

28) Match the column-I with column-II :

Column-I		**Column-II**	
(i)	Complex-I of ETS	(a)	FAD, Fe-S
(ii)	Complex-II of ETS	(b)	FMN, Fe-S
(iii)	Complex-III of ETS	(c)	Cytb-$cytc_1$, Fe-S
(iv)	Complex-IV of ETS	(d)	Cyt a & $Cyta_3$, 2 Cu centres

(1) i-b, ii-a, iii-c, iv-d

(2) i-a, ii-b, iii-c, iv-d

(3) i-d, ii-c, iii-b, iv-a

(4) i-a, ii-c, iii-b, iv-d

29) Match the column I and column II and choose the correct option –

Column - I		Column - II	
A.	Pyruvate dehydrogenase	(i)	2PGA → PEP
B.	Pyruvic acid kinase	(ii)	1,3 BiPGA → 3PGA
C.	Enolase	(iii)	Pyruvic acid → Acetyl CoA
D.	Phosphoglycerate kinase	(iv)	PEP → Pyruvic acid

(1) A – (ii), B – (iv), C – (i), D – (iii)

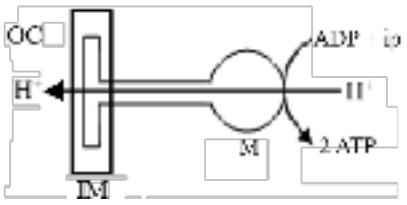
(2) A – (iii), B – (ii), C – (i), D – (ii)

(3) A – (iii), B – (i), C – (iv), D – (ii)

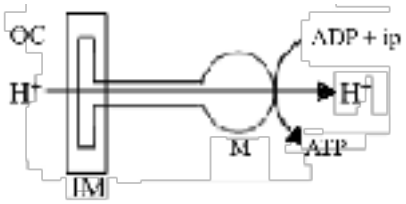
(4) A – (iii), B – (iv), C – (i), D – (ii)

30) Which diagram represent the ATP synthesis in mitochondria through chemi-osmosis?
I.M = Inner membrane
M = Matrix
O.C = outer chamber

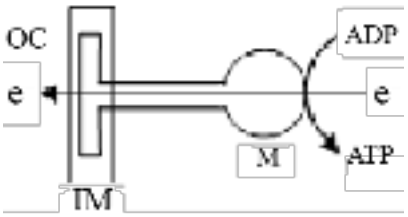
(1)



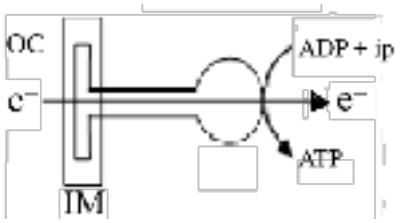
(2)



(3)



(4)



31) Growth is expressed as :

(1) increase in fresh weight

(2) increase in dry weight

(3) increase in surface area

(4) All of these

32) Heterophylly is an example of :-

(1) Growth
(2) Abscision
(3) senescence
(4) Plasticity

33) Which of the following is incorrectly match :-

(1) Ethylene - Cousins
(2) Kinetin = Skoog and Miller
(3) Gibberellic acid = F.W. Went
(4) Auxin - Darwin

34) Which of the given hormone is used to produce parthenocorpic or seedless tomatoes ?

(1) Auxin
(2) Cytokinin
(3) Ethylene
(4) Gibberellin

35) Which PGR (Plant growth regulator) is used to speed up the malting process in brewing industry :-

(1) Auxin
(2) Cytokinin
(3) Gibberellin
(4) Ethylene

36) Which of the following hormone promote new leaves and adventitious shoot formation ?

(1) Auxin
(2) Gibberellins
(3) Cytokinin
(4) Ethylene

37) Which growth regulator promotes root growth and root hair formation, thus helping the plant to increase their absorption surface :-

(1) Kinetin
(2) Ethylene
(3) Auxin
(4) Gibberellin

38) Choose the incorrect statement from the following.

(1) A sigmoid curve is a characteristic of living organism growing in a natural environment.

(2) The growth of a pollen tube is measured in terms of its length

(3) In geometric growth only one daughter cell continues to divide while the other differentiates and matures.

(4) Nutrients are required by plant for the synthesis of protoplasm and act as source of energy

39) Choose the incorrect statement :

(1) Abscisic acid was discovered for its role in regulating abscission and dormancy

(2) Abscissic acid plays an important role in seed development, maturation & dormancy

(3) Abscisic acid helps seeds to withstand dessication and other factors unfavorable for growth

(4) ABA acts as a general plant growth promotor and promoter of plant metabolism

40) Match the column-I with column-II.

	Column-I		**Column-II**
(A)	Plant growth	(i)	Cells attains maximal size [In terms of wall thickning and protoplasmic modification]
(B)	Phase of maturation	(ii)	17,500 cells / hour
(C)	Single maize root apical meristem	(iii)	Open & generally indeterminate

(1) A-iii, B-ii, C-i

(2) A-iii, B-i, C-ii

(3) A-i, B-ii, C-iii

(4) A-ii, B-i, C-iii

41) Match the column-I with the column-II :-

	Column-I		**Column-II**
(A)	Formation of cork cambium	(I)	Differentiation
(B)	Primary meristmetic cells has lost the capacity of division	(II)	Redifferentiation
(C)	Cells formed from dedifferentiation and lost the capacity of division.	(III)	Dedifferentiation.

(1) A-II, B-I, C-III

(2) A-I, B-II, C-III

(3) A-III, B-I, C-II

(4) A-I, B-III, C-II

42) Match the following column I and column II and select the correct option given below :

Column I	**Column II**

a.	Auxin	i.	Cytokinin
b.	Zeatin	ii	Stress hormone
c.	ABA	iii	Fruit ripener
d.	Ethephon	iv	Apical dominance

(1) a-i,b-ii,c-iii,d-iv

(2) a-iv,b-i,c-iii,d-ii

(3) a-iv,b-i,c-ii,d-iii

(4) a-ii,b-i,c-iii,d-iv

43) The volatile substance is released from ripened oranges that __A__ the ripening of stored un-ripened banana. It was identified as __B__.

(1) A-Reduced , B-Ascisic acid

(2) A-Hastened, B-Ethylene

(3) A-Stop, B-Auxin

(4) A-Hastened, B-Cytokinins

44) Match List-I with List-II –

List-I		**List-II**	
(a)	Developmental Heterophylly	(I)	Growth and Differentiation
(b)	Environmental Heterophylly	(II)	Cotton
(c)	Development	(III)	Buttercup

Choose the correct answer form the options given below:

(1) (a) – (II), (b) – (III), (c) – (I)

(2) (a) – (III), (b) – (II), (c) – (I)

(3) (a) – (I), (b) – (II), (c) – (III)

(4) (a) – (III), (b) – (I), (c) – (II)

45) Given below two statements.
Statement-I : Development is a term that includes all changes that an organism goes through during its life cycle from germination of seed to senescence.
Statement-II : Gibberellins, cause fruits like apple to elongate and improve its shape.
Choose the correct answer from the options given below.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are incorrect.

46) Match the following and choose the correct of option.

Column-I		Column-II	
(a)	Thalamus	(i)	Hunger, thirst
(b)	Hypothalamus	(ii)	Relay centre
(c)	Cerebellum	(iii)	Balance and posture
(d)	Medulla	(iv)	Cardiac and respiratory centres
		(v)	Association areas
		(vi)	Emotional control

(1) a-(ii), b-(i), c-(iii), d-(iv)

(2) a-(i), b-(ii), c-(iv), d-(iii)

(3) a-(ii), b-(iii), c-(i), d-(iv)

(4) a-(iii), b-(iv), c-(ii), d-(i)

47) **Assertion (A) :** Cerebral aqueduct passes through the midbrain.
Reason (R) : Midbrain is located between thalamus/hypothalamus and pons.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

48) Match the List-I with List-II.

	List-I		List-II
(1)	Myelinated nerve fibres	(a)	Somatic neural system
(2)	Unmyelinated nerve fibres	(b)	Embryonic stage
(3)	Unipolar	(c)	Cerebral cortex
(4)	Multipolar	(d)	Spinal & Cranial nerve

(1) 1-a, 2-b, 3-c, 4-d

(2) 1-b, 2-c, 3-a, 4-d

(3) 1-d, 2-b, 3-c, 4-a

(4) 1-d, 2-a, 3-b, 4-c

49) **Statement-I :** Neurotransmitters are involved in the transmission of impulses at electrical synapses.
Statement-II : Transmission of an impulse across electrical synapse is very similar to impulse conduction along a single axon.

(1) Both statements I & II are correct.

(2) Only statement I is correct.

(3) Only statement II is correct.

(4) Both statements are incorrect.

50) Choose the incorrect statements from given below options.

(1) Coordination is the process through two or more organs interact and complement the functions of each other.

(2) Neural system provides an organised network of neuron for quick coordination.

(3) Neural system provides point to point connection for quick coordination.

(4) In our body only neural system coordinates and integrates the all the activites of organs.

51) Part of the peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system is -

(1) Parasympathetic neural system.

(2) Sympathetic neural system

(3) Autonomic neural system

(4) Visceral Nervous system

52) When a neuron is not conducting any impulse, the axon membrane is :-

(1) Equally permeable for both K^+ and negatively charged proteins.

(2) Comparatively more permeable to K^+ ions and nearly impermeable to negatively charged proteins.

(3) Comparatively more permeable to Na^+ ions and nearly impermeable to K^+ ion.

(4) Nearly impermeable to Na^+ ion and impermeable to negatively charged proteins.

53) A chemical synapse the membrane of pre and post synaptic neurons is separated by fluid filled spaces known as :-

(1) Synaptic vesicle

(2) Synaptic knob

(3) Synaptic cleft

(4) Both (1) & (3)

54) What happens during the repolarisation phase of neuron?

(1) Influx of Na^+ continues

(2) Eflux of K^+ occurs

(3) Ca^{2+} influx occurs

(4) Na^+/K^+ pump is inactive

55) The brain is covered by __(a)__ inside the skull. The human brain is well protected by the __(b)__.
Find the correct option which fills above blanks correctly :-

(1) (a) - Adipose tissue,
(b) - Epineurium

(2) (a) - Connective tissue,
(b) - Epimysium

(3) (a) - Cranial meninges,
(b) - Skull

(4) (a) - Vertebral column,
(b) - Cranial meninges

56) Which areas of the human brain are chiefly responsible for managing emotional reactions (such as fear, anger, pleasure) and regulation of sexual behavious ?

(1) Amygdala and medulla oblongata

(2) Limbic system and hypothalamus

(3) Cerebellum

(4) Pituitary gland

57) Which part of the hind brain interconnect different regions of the brain-

(1) Pons

(2) Medulla

(3) Cerebellum

(4) Cerebral aqueduct

58) Select the correct matching:

	Column-I		**Column-II**		**Column-III**
A.	Paramecium	1.	Cilia	X.	Movement of food in cytopharynx and locomotion.
B.	Hydra	2.	Tentacles	Y.	Capturing of prey and for locomotion.
C.	Human	3.	Limbs	Z.	Changes in body posture and for locomotion.

(1) A−1−X, B−2−Y, C−3−Z

(2) A−1−Z, B−3−X, C−2−Y

(3) A−3−Z, B−1−X, C−2−Y

(4) A−2−Y, B−3−X, C−1−Z

59) How many of the following bones are paired.
(Parietal, temporal, sphenoid, ethmoid, Nasal, Maxilla, Mandible, Hyoid, Lacrimal)

(1) 4

(2) 3

(3) 5

(4) 6

60) **Assertion (A) :** Joints are points of contact between bones or between bones and cartilages.
Reason (R) : Force generated by the muscles is used to carry out movement through joint.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

61) Given below are two statements : one is labelled as Assertion 'A' and other as Reason 'R'.
Assertion (A) : Sarcoplasmic reticulum of the muscle fibre is the store house of calcium ions.
Reason (R) : Muscle fibre is a syncytium as the sarcoplasm contains many nucleus.
in the light of the above statements, choose the correct answer from the options given below :

(1) A and R both statement are correct, R is correct explanation of 'A'.

(2) A and R both statement are correct, R is not correct explanation of 'A'.

(3) A is correct but R is false.

(4) A is false but R is true.

62) Given below are two statements :
Statement-I : Smooth muscles have a striped appearance.
Statement-II : Each actin filament has two 'F' actins helically wound to each other.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statemen-II is incorrect.

(3) Statement-I is incorrect but Statemen-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

63) **Statement-I :** Each rib is a thin flat bone connected dorsally to the sternum and ventrally to the vertebral column.
Statement-II : Ribs has two articulation surfaces on its dorsal end and is hence called bicephalic.

(1) Statement-I is correct Statement-II is incorrect.

(2) Statement-I is incorrect Statement-II is correct.

(3) Both Statement-I and Statement-I is correct.

(4) Both Statement-I and Statement-I is incorrect.

64) Which one of the following statements is false ?

(1) Clavicle bone articulates with acromian process of scapula.

(2) Head of humerus bone articulates with glenoid cavity.

(3) The skull is composed of two sets of bone carpal and facial.

(4) The vertebral column protects the spinal cord.

65) Select the correct statements
(A) actin are pulled inwards thereby causing a shortening of the sarcomere.

(B) during muscle contraction 'A' bands retain the length
(C) The Ca^{++} ions are pumped back to the sarcoplasmic cisternae resulting in the demasking of actin filament.
(D) Repeated activation of the muscle can lead to the accumulation of lactic acid.
Choose the most appropiate answer from the option give below

(1) B, C and D only
(2) C and D only
(3) A, B and D only
(4) B and D only

66) In the resting state, a submit of troponin masks the active binding sites for myosin on :

(1) Meromyosin filament
(2) Actin filament
(3) Tropomyosin
(4) M-line

67) Progressive degeneration of skeletal muscles mostly due to genetic disorder is :-

(1) Myesthenia gravis
(2) Muscular dystrophy
(3) Gout
(4) Tetany

68) What is the red coloured oxygen storing pigment found in muscles ?

(1) Cytochrome
(2) Chlorophyll
(3) Haemoglobin
(4) Myoglobin

69) Which of the following statement is not correct about cardiac muscle?

(1) Cardiac muscle only found in heart
(2) Cardiac muscle are involuntary in nature
(3) Cardiac muscle are non striated
(4) Cardiac muscle not controlled by nervous system directly.

70) Which disorder may occurrs in female after menopause ?

(1) Tetany
(2) Osteoporosis
(3) Rheumatoid arthritis
(4) Myasthenia gravis

71) Scapula is large triangular flate bone situated in the dorsal part of the thorax between

(1) Second and seventh ribs

(2) Second and fifth ribs

(3) Third and Eight ribs

(4) Fourth and seventh ribs

72) Below given in diagram of hind limb of human. 

Identify 'a' and choose the correct answer.

(1) Radius

(2) Tibia

(3) Ulna

(4) Fibula

73)

Which is a correct match regarding both forelimbs?

Column - I		Column - II	
(1)	Carpals	(p)	16
(2)	Metacarpals	(q)	5
(3)	Phalanges	(r)	24
(4)	Radius	(s)	1

(1) 1

(2) 2

(3) 3

(4) 4

74) Pituitary → TSH → Thyroid → T_3/T_4 → _____
Which Statements are true for the above blank space ?
I. T_3/T_4 increase RBC production
II. T_3/T_4 require iodine for synthesis
Ill. Hyperthyroidism causes weight gain
IV. T_3/T_4 regulate metabolism of carbohydrates, proteins and fats

Choose the correct option related to above informations :

(1) I, II, IV

(2) II, IV only

(3) I, II, III

(4) I and III only

75) The hypothalamus controls the activity of both anterior and posterior parts of the pituitary gland. It does so through hormonal and neural mechanisms. The anterior pituitary is regulated via a portal system while the posterior pituitary is under direct neural control.
Statement-I : The hypothalamus secretes oxytocin and prolactin which are stored in the posterior pituitary.
Statement-II : The anterior pituitary is regulated by releasing and inhibiting hormones of hypothalamus through a portal circulation.

(1) Both statement-I and statement-II are true.

(2) Statement-I is true, Statement-II is false.

(3) Statement-I is false, Statement-II is true.

(4) Both statement-I and statement-II are false.

76) **Assertion :** The pineal gland is located on the dorsal side of hind-brain.
Reason : Melatonin plays a very important role in the regulation of a diurnal rhythm of our body.

(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

77) Choose the correct answer from the options given below about mode of action of the hormone :-

(1) Steroidal hormone do not enter the target cell but intract with cell membrane only.

(2) Hormone which interacts with intra cellular receptors are proteinaceous hormone.

(3) Protein nature hormone generates the secondary messanger after binding cell membrane.

(4) After generation of secondary messanger biochemical response occurs in cytoplasm of cell by steroid hormone.

78) **Statement-1 :** Over secretion of GH stimulates abnormal growth of the body leading to gigantism.
Statement-2 : Prolactin regulates the growth of the mammary gland.

(1) Both statement are correct.

(2) Both statement are incorrect.

(3) Only statement 1 is correct.

(4) Only statement 2 is correct.

79)

Match the Column I with Column II and choose the correct option :-

Column I		Column II	
(A)	Aldosterone	(i)	Piloerection
(B)	Cortisol	(ii)	Increase BP
(C)	Adrenaline	(iii)	Growth of axillary hair
(D)	Androgenic steroids	(iv)	Increase blood glucose

	(A)	(B)	(C)	(D)
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(iii)	(i)	(iv)
(3)	(ii)	(iv)	(i)	(iii)
(4)	(i)	(iii)	(iv)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

80) Match the List-I with List-II

List-I		List-II	
A	Heart	I	Atrial natriuretic factor
B	Liver	II	Erythropoietin
C	Kidney	III	Angiotensinogen
D	GIT	IV	CCK

Choose the correct answer from the options given below

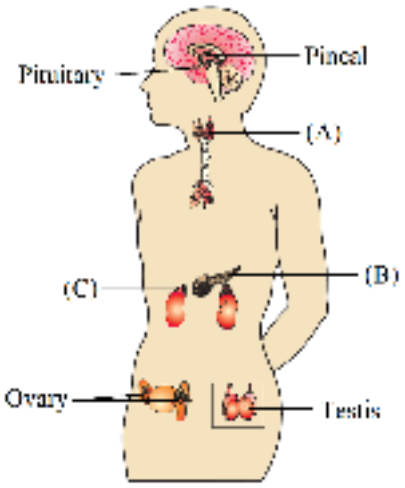
(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-III, D-IV

(3) A-III, B-II, C-I, D-IV

(4) A-I, B-III, C-II, D-IV

81)



	(A)	(B)	(C)
1.	Thyroid	Parathyroid	Adrenal
2.	Parathyroid	Liver	Pancreas
3.	Thyroid	Pancreas	Adrenal
4.	Thyroid	Pancreas	Thymus

(1) 1

(2) 2

(3) 3

(4) 4

82)

Choose the incorrect for current scientific definition of hormones :

(1) Non nutrient chemicals

(2) Intercellular messengers

(3) Produced is trace amounts

(4) Produced by endocrine glands and released into the blood and saliva

83) Which of the following conditions will decrease release of parathyroid hormone from parathyroid gland ?

(1) Rise in blood Ca^{++} level

(2) Fall in blood Ca^{++} level

(3) Rise in bone Ca^{++} level

(4) Fall in active vitamin D levels.

84) Immunity gets weaker in old aged person due to

(1) High secretion of thymosin

(2) Thymus is degenerated

(3) Deficiency of Insulin

(4) Hyper secretion of Thyroxin

85) Which of the following hormones lowers the blood glucose level?

(1) Glucagon
(2) Insulin
(3) Cortisol
(4) Growth hormone

86) Which of the following is not a function of Testosterone ?

(1) Development and maturation of male accessory sex organs.
(2) Appearance of male secondary sexual characters.
(3) Regulates male sexual behaviour, anabolic pathways and erythropoiesis.
(4) Formation of sac like structures and stimulates milk secretion.

87) Estrogen is synthesised and secreted mainly by :-

(1) Stromal tissues
(2) Growing ovarian follicle
(3) Corpus luteum
(4) Ruptured graafian follicle

88) Fill the blanks by selecting correct option.
MSH acts on the ___A___and regulates___B___.

(1) A–Gonads, B–Gametogenesis
(2) A–Melanocytes, B–Pigmentation
(3) A–Bones, B–Growth
(4) A–Ovary, B–Follicular growth

89) Protrusion of the eye ball, increased basal metabolic rate and weight loss are characteristics of which disease ?

(1) Diabetes mellitus
(2) Graves disease
(3) Addison disease
(4) Caushing syndrom

90) Which of the following increases the strength of heart contraction ?

(1) Thymosin
(2) ACTH
(3) Adrenaline
(4) Melatonin

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	2	4	1	3	4	4	4	1	3	1	3	1	4	4	2	1	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	2	4	2	4	4	3	1	2	4	1	1	3	4	1	4	2	4	3
Q.	41	42	43	44	45															
A.	3	1	4	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	2	1	4	3	1	2	1	4	2	4	3	2	3	4	2	1	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	1	2	4	4	2	4	2	3	2	3	4	3	1	1	4	3	4	1
Q.	86	87	88	89	90															
A.	4	1	3	4	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	3	2	3	1	2	2	4	2	1	3	1	3	4	4	2	4	1	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	3	1	2	2	1	1	4	2	4	4	3	1	3	3	2	3	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	2	1	1	1	2	4	3	4	4	4	3	2	3	2	1	1	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	3	2	3	3	2	2	4	3	2	1	2	1	1	3	4	3	1	3	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	4	1	2	2	4	2	2	2	3										

SOLUTIONS

PHYSICS

1) Force F = – ax – b can be rewritten as $F = -a\left(x + \frac{b}{a}\right)$, indicating oscillation about x = $-\frac{b}{a}$, Hence the motion is SHM . Statement -1 is false and statement -2 correctly describes the condition for SHM.

2) • Effective Spring Constant for Parallel Springs:
$k_{eff} = k_1 + k_2$
• Time Period:
$T = 2\pi\sqrt{\frac{m}{k_{eff}}}$
The stiffness of a spring is inversely proportional to its length. Therefore the stiffness of each part is
$k \propto \frac{1}{\ell}$
$k_1 = \frac{4}{3}k$ and $k_2 = 4k$
Time period $T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$
$= 2\pi\sqrt{\frac{3m}{16k}} = \frac{\pi}{2}\sqrt{\frac{3m}{k}}$

3) $\frac{\Delta\phi}{2\pi} = \frac{\Delta t}{T}$



Phase diff. $\Delta\phi = \frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$
time taken, $t = \frac{T}{2\pi}\Delta\phi$
$t = \frac{T}{2\pi} \times \frac{7\pi}{12}$
$t = \frac{7T}{24}$

4) $A_{net} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$
$TE = \frac{1}{2}KA^2 = \frac{1}{2} \times 100 \times \left(4\sqrt{2}\right)^2$
$= 50 \times 16 \times 2 = 1600$ J

5)

$v = \omega\sqrt{A^2 - x^2}$

(Relation between velocity & position is)

Original speed at that x'

$v = \omega\sqrt{A^2 - x^2}$ ....(1)

After tripling the speed

$v' = 3v = \omega\sqrt{(A_{new})^2 - x^2}$ ....(2)

Squaring equation (2) and putting the value from equation (1)

$\Rightarrow 9v^2 = \omega^2\left[(A_{new})^2 - (x)^2\right]$

$\Rightarrow 9 = \dfrac{\omega^2\left[(A_{new})^2 - \left(x^2\right)\right]}{\omega^2\left(a^2 - x^2\right)}$

$\Rightarrow 9\left[A^2 - x^2\right] = \left[(A_{new})^2 - (x)^2\right]$

putting $x = \dfrac{2A}{3}$

$A_{new} = \dfrac{7A}{3}$

6)

General equation of a sinusoidal wave is

$y = A \sin(\omega t - kx + \phi)$

$v = A \sin\left(\omega t - kx + \phi + \dfrac{\pi}{2}\right)$

$a = A \sin(\omega t - kx + \phi + \pi)$

7) $K = \dfrac{F}{x} = \dfrac{8}{2} = 4$ N/m

$T = 2\pi\sqrt{\dfrac{m}{K}} = 2\pi\sqrt{\dfrac{0.01}{4}}$

$= 0.3$ sec

8) $x = \sin\left(\dfrac{2\pi}{8}t\right) = \sin\left(\dfrac{\pi}{4}t\right)$

$\alpha = -\omega^2 x = -\dfrac{\pi}{4} \times \dfrac{\pi}{4} \times \sin\left(\dfrac{\pi}{4} \times \dfrac{4}{3}\right)$

$= -\dfrac{\pi^2}{16} \times \dfrac{\sqrt{3}}{2} = \dfrac{-\sqrt{3}\pi^2}{32}$ cm/s^2

9)

$T = 2\pi\sqrt{\dfrac{\ell}{g}}$

$$T' = 2\pi\sqrt{\frac{1.21\ell}{g}} = 1.1\left(2\pi\sqrt{\frac{\ell}{g}}\right) = 1.1\ T$$

⇒ % increase in time period will be 10%

10) Beat frequency = number of beats/sec,
⇒ $n = |n_2 - n_1|$
$\therefore\ n_1 = n_2 \pm n.$

11)

$f_{oop} = 3f_{cop} \Rightarrow \dfrac{v}{2\ell_{oop}} = \dfrac{3v}{4\ell_{cop}}$

$\Rightarrow \ell_{oop} = \dfrac{2}{3} \times \ell_{cop} = \dfrac{2}{3} \times 20 = 13.3$ cm

12) $I_1 = I + I + 2\sqrt{I}\sqrt{I}\cos 0^\circ \Rightarrow 4I$

$I_2 = I + I + 2\sqrt{I}\sqrt{I}\cos 90^\circ \Rightarrow 2I$

$\dfrac{I_1}{I_2} = \dfrac{2}{1}$

13)

Wave speed in a string depends on square root of tension and mass per unit length (μ)
Formula :-

$v \propto \sqrt{\dfrac{T}{\mu}}$

$\dfrac{v_2}{v_1} = \sqrt{\dfrac{T_2}{T_1}}$

Calculation :-

$v_2 = v_1 + \dfrac{20}{100} \times 150$

$v_2 = 180$ m/s

$\dfrac{180}{150} = \sqrt{\dfrac{T_2}{120}} \Rightarrow \dfrac{6}{5} = \sqrt{\dfrac{T_2}{120}}$

squaring : $\dfrac{36}{25} = \dfrac{T_2}{120} \Rightarrow T_2 = \dfrac{36}{25} \times 120 = 172.8$

Increase = $\dfrac{172.8 - 120}{120} \times 100 = 44\%$

14) In a stationary wave antinodes are the points where the displacement is maximum because the particle oscillate with the greatest amplitude and at these point pressure change in minimum.

15) Maximum particle velocity = Aω

wave velocity = $\frac{\omega}{K}$

$\frac{Vp_{max}}{V_w} = AK.$

in equation

$y = 3 \sin \frac{\pi}{2}(50t - x)$

$y = 3 \sin \left(25\pi t - \frac{\pi}{2}x\right)$

Compare it with

$y = A \sin (\omega t - kx)$

$AK = \frac{3\pi}{2}$

16)

Progressive wave on string (Power and Intensity)

$\because I \propto (a)^2 \cdot (\omega)^2$

For y_1:

$I_1 \propto (2a)^2\omega^2 = 4a^2\omega^2$

For y_2 :

$I_2 \propto (a)^2 (2\omega)^2 = a^2 \times 4\omega^2 = 4a^2\omega^2$

Since $I_1 = I_2$

17) The plucking point of the string will be the antinode and touching point will be node.

18) $T = 2\sqrt{\frac{L}{g}}$

$T = 2\sqrt{\frac{L}{g}}$ ....(1)

$T' = 2\sqrt{\frac{L}{2g}}$ ....(2)

$T' = \frac{T}{\sqrt{2}}$

19) Beat frequency

$f_{beat} = |f_1 - f_2|$

$f_1 = 200$ Hz



$f_2 = 196$ Hz

Adding wax. f↓

$f_2 = 195$ Hz (5 b/s)

$f_2 = 194$ Hz (6 b/s)

$f_2 = 204$ Hz

Adding wax. f↓

$f_2 = 203$ Hz (3 b/s)

$f_2 = 202$ Hz (2 b/s)

20) $\frac{\Delta\phi}{2\pi} = \frac{\Delta x}{\lambda} = \frac{\Delta t}{T}$

$\frac{\Delta x}{\lambda} = \frac{\Delta t}{T}$

from the given equation $\omega = 2\pi = \frac{2\pi}{T}$

T = 1 sec.

K = $\frac{2\pi}{\lambda} = \frac{\pi}{2} \Rightarrow \lambda = 4m.$

Now $\frac{5-2}{4} = \Delta t$

$\Delta t = \frac{3}{4}$ sec.

21) y = A cos(ωt – kx) & $y = e^{-(x-vt)^2}$

are function which are finite and can be written as f(ax ± bt), thus can represent a wave -

22) $\frac{A}{2} = Ae^{\frac{-bt}{2m}}$

$\frac{A}{x} = Ae^{\left(\frac{-bt}{2m}\right)\times 3} \Rightarrow x = 8$

23) Heat lost = Heat gain

$ms_1 (32 – 20) = ms_2 (40 – 32)$

$12s_1 = 8s_2$

$\frac{s_1}{s_2} = \frac{2}{3}$

24) $\ell_1 + e = \frac{\lambda}{4} = \frac{V}{4n} = \frac{336}{4 \times 504}$

Here e = 0.3 D

e = 0.3 × 6 = 1.8 cm

$\ell_1 + 1.8 = 16.6$

$\ell_1 = 14.8$ cm

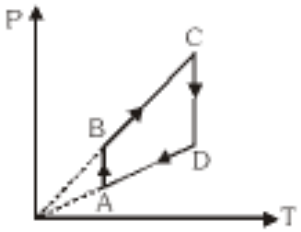
25) For any scale

$\frac{x - LFP}{UFP - LFP} = \text{const.}$

$\frac{\frac{x_0}{2} - \frac{x_0}{3}}{x_0 - x_0/3} = \frac{C-0}{100-0}$

$\frac{\frac{x_0}{6}}{\frac{2x_0}{3}} = \frac{C}{100}$

C = 25°C



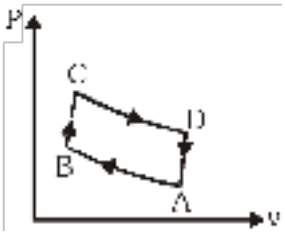
26)

AB → T = Const.

BC → v = Const.

CD → T = Const.

DA → V = Const.



27) Since one of the spheres is solid and the other hollow, of equal radii, the masses of the two spheres are different. The radiating surface of the two spheres are the same. The cooling rates (i.e.,) the fall of temperature per second will NOT be the same for the two spheres as the masses are different.

$\boxed{R_C \propto \frac{R_H}{mass}}$ or $R_C \propto \frac{1}{mass}$

R_C = Rate of cooling

R_H = Rate of loss of heat

28) For series combination,

$k_s = \frac{2k_1k_2}{k_1 + k_2} = 2\frac{2.k.2k}{k + 2k}$

$k_s = \frac{4k}{3}$

For parallel combination,

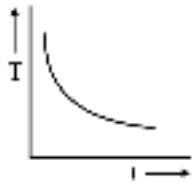
$k_p = \frac{k_1 + k_2}{2} = \frac{K + 2k}{2}$

$k_p = \frac{3k}{2}$

$\frac{k_s}{k_p} = \frac{4k}{3} \times \frac{2}{3k} = \frac{8}{9}$

29)

A/C to Newton's law of cooling



30) $T_{mix} = \dfrac{m_1 s_1 T_1 + m_2 s_2 T_2}{m_1 s_1 + m_2 s_2} = \dfrac{2m\frac{C}{2}T + \frac{m}{3}2C\,3T}{2m\frac{C}{2} + \frac{m}{3}.2C}$

$T_{mix} = \dfrac{9T}{5}$

31) ∵ PV = nRT

$\dfrac{P_1}{P_2} = \dfrac{m_1}{m_2} \times \dfrac{T_1}{T_2}$

$= \dfrac{P}{2} = \dfrac{6}{x} \times \dfrac{500}{300}$

x = 5 gm

So, leaked amount ⇒ 6 – 5 = 1g

32)

PV = nRT

Slope = $\dfrac{T}{P} = \dfrac{V}{nR} \propto \dfrac{1}{n} \propto \dfrac{1}{m}$

$(Slope)_2 > (Slope)_1$

$m_1 > m_2$

33)

According to first law

Q = ΔU + WD

$5 = 0 + (WD)_{AB} + (WD)_{BC} + (WD)_{CA}$

$5 = 10 + 0 + (WD)_{CA}$

$(WD)_{CA} = 5 - 10 = -5J$

34)

(i) $P \propto \dfrac{1}{V} \Rightarrow$ T = const.

(ii) $P \propto V$

(iii) P = const.

(iv) V = const.

35)

$Ms\Delta t = mL_f$

$4 \times 500 \times (425 - 0) = m(336) \times 10^3$

m = 2.529 kg.

36)

$\gamma_{mix} = 1 + \frac{2}{f_{mix}}$

$f_{mix} = \frac{\mu_1 f_1 + \mu_2 f_2}{\mu_1 + \mu_2} = \frac{1 \times 3 + 1 \times 5}{1 + 1}$

$f_{mix} = 4$, so, $\gamma_{mix} = 1 + \frac{2}{4}$

$\gamma_{mix} = \frac{3}{2}$

37) In ΔOAB,

$OA^2 = AB^2 - OB^2$

Before heating,

$OA^2 = \ell^2 - \frac{\ell^2}{4}$ ...(1)

After heating,

$OA^2 = [\ell\{1 + \alpha_2.t\}]^2 - \left[\frac{\ell}{2}\{1 + \alpha_1 t\}\right]^2$ ...(2)

$OA^2 = \square^2 (1 + 2\alpha_2 t) - \frac{\ell^2}{4} (1 + 2\alpha_1 t)$

$\ell^2 - \frac{\ell^2}{4} = \ell^2 (1 + 2\alpha_2 t) - \frac{\ell^2}{4} (1 + 2\alpha_1 t)$

on solving we get

$(\alpha_1 = 4\alpha_2)$

38) $\lambda_m \propto \frac{1}{T}$ (wein's law)

$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$

$\lambda_2 = \frac{\lambda_1 T_1}{T_2}$

$\lambda_2 = \frac{6000 \times 4800}{3000}$

$\lambda_2 = 9600$ Å

39) $P = aT^{-1}$

$PT = a$ = constant

$P\left[\frac{PV}{nR}\right]$ = constant $\Rightarrow PV^{+1/2}$ = constant

$W = \frac{nR\Delta T}{1 - x} = \frac{nR(4T - T)}{1 - \frac{1}{2}} = 6nRT$

40) 

Let T' be the temperature of the middle plate (II) and A be area of each

plate. Under steady state, the rate of energy received by the middle plate is equal to rate of energy emitted by it.

i.e. $\sigma A(3T)^4 - \sigma A(T')^4 = \sigma A(T')^4 - \sigma A(2T)^4$

or $\sigma A [(3T)^4 - (T')^4] = \sigma A [(T')^4 - (2T)^4]$

$(3T')^4 - (T')^4 = (T')^4 - (2T')^4$

or $2(T')^4 = T^4(3^4 + 2^4) = T^4(81 + 16) = 97T^4$

or $T'^4 = \frac{97}{2}T^4$

or $T' = \left(\frac{97}{2}\right)^{1/4} T$

41) For adiabatic expansion

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$\Rightarrow T_2 = \frac{300}{(2)^{1.5-1}} = 212K$

Now,

$dU = dW = \frac{R}{\gamma-1}(T_1 - T_2)$

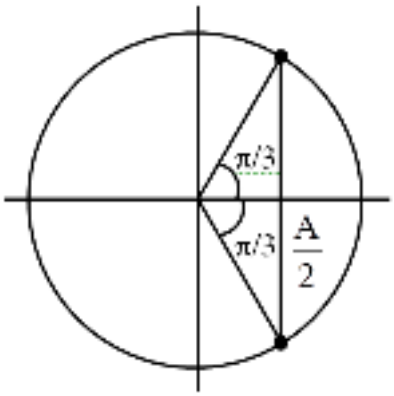
$\Rightarrow \frac{R}{(1.5-1)}(300-212)$

= 1462 J

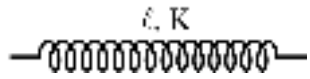
42) $\frac{70-50}{10} = k\left(\frac{70+50}{2} - 30\right)$ ....(1)

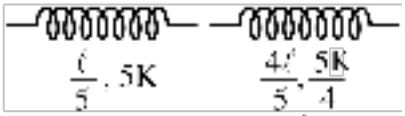
$\frac{50-40}{t} = k\left(\frac{50+40}{2} - 30\right)$ .....(2)

 t = 10 min.

43) 

Phase difference = $\frac{2\pi}{3}$

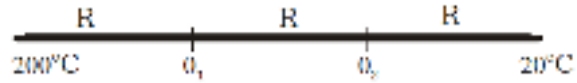
44) 



as force constant $K \propto \frac{1}{\ell}$



45) Resistance of all identical rods are same.



$H_1 = H_2 = H_3$

$$\frac{KA(200-\theta_1)}{\ell} = \frac{KA(\theta_1-\theta_2)}{\ell} = \frac{KA(\theta_2-20)}{\ell}$$

$\theta_1 = 140°C$ $\theta_2 = 80°C$

CHEMISTRY

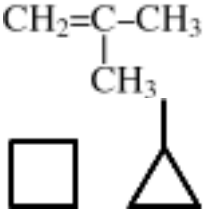
46)

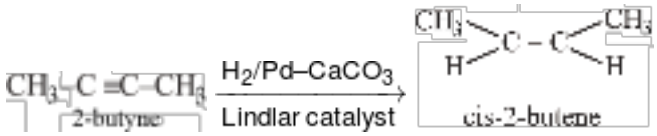
Carbanion stability is increased by electron-withdrawing groups (-M, -I) and decreased by electron-donating groups (+M, +H, +I).

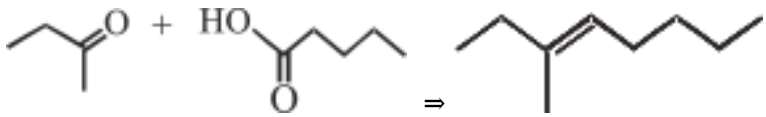
Cation	Group	Dominant Effect	Stability Rank
(I)	$-OCH_3$	Strong +M (destabilizing)	4 (**Least Stable**)
(II)	$-NO_2$	**Strong -M** (stabilizing)	1 (**Most Stable**)
(III)	–H	Reference (Resonance)	2
(IV)	$-CH_3$	+H / +I (destabilizing)	3

The **Correct** decreasing order of stability is (II) > (III) > (IV) > (I).
Final answer :- (2).

47) $CH_2=CH-CH_2-CH_3$
$CH_3-CH=CH-CH_3$ (Cis & trans)



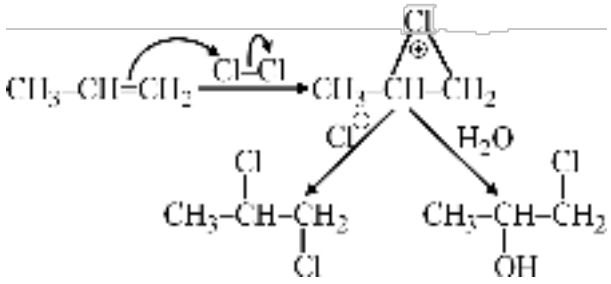
48) 

49) 

50)

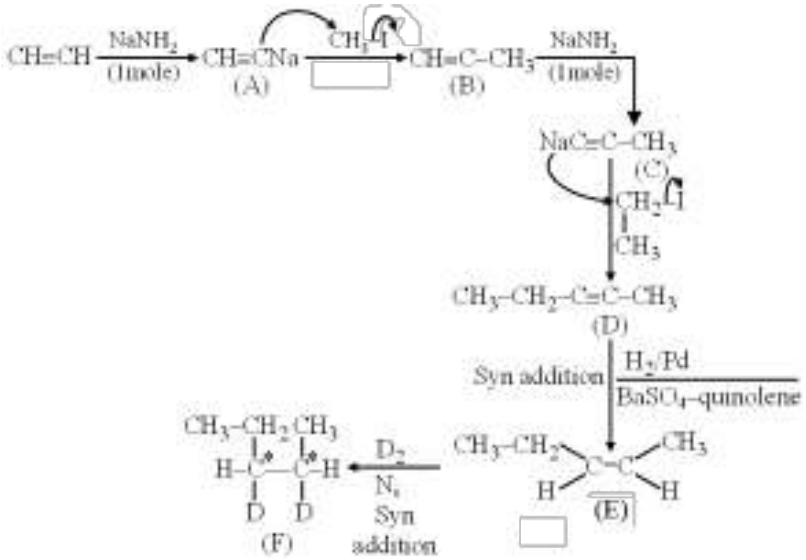
$$CH_3\text{–}CH{=}CH_2 \xrightarrow{Br_2} CH_3\text{–}\underset{\displaystyle Br}{\underset{|}{CH}}\text{–}CH_2Br$$

$$CH_3\text{–}\underset{\displaystyle Br}{\underset{|}{CH}}\text{–}CH_2Br \xrightarrow[\text{Halide Exchange}\ (-Br^-)]{\overset{\ominus}{Cl}} CH_3\text{–}\underset{\displaystyle Cl}{\underset{|}{CH}}\text{–}CH_2\text{–}Br$$

51) 

$CH_3\text{–}\underset{\displaystyle OH}{\underset{|}{CH}}\text{–}\underset{\displaystyle OH}{\underset{|}{CH_2}}$ does not form in given reaction.

52) **Question Explanation**: Identifying the incorrect statement.
Concept: Basicity and aromaticity
Solution: The statement that B is more basic than A is correct.
Final Answer: 1



53)

F can not be meso it should be chiral

54) PCC have 5C including multiple bonds. So we will write word root 'pent'. Numbering will start from right side. So, that multiple bonds gets lowest number. Triple bond at position 1. So, suffix 1-yne will used.

Double bond at position 3. So suffix 3-ene will used.

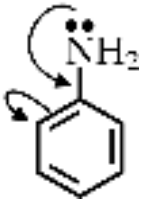
'en' and 'yne' are written in alphabatical order. So, the IUPAC name of given compound is : Pent-3-en-1-yne

Hence,

Option (1) is correct : Pent-3-en-1-yne

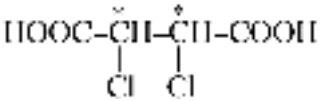
55)

A. Resonance involves the delocalization of a lone pair or π electrons. In options A and B, the lone pair on the nitrogen atom is not conjugated with a π system, so it is not involved in resonance. In option C, the lone pair on the nitrogen atom is not conjugated. In option D, the lone pair on the nitrogen atom is conjugated with the π system of the benzene ring.
B. Lone pair in conjugation with π bond is in resonance.



Answer 4

A. Lone pair on π bonded atom is not in resonance.



56)

shows optical isomerism.

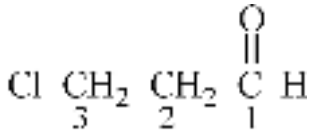
57) **Solution:**

→ Anti form of conformer is more stable than gauche form.

→ Minimum the energy more is the stability
→ Anti form of n-butane has minimum energy.

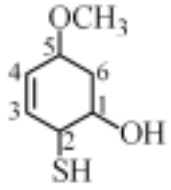
Hence,
Correct option: (4) Energy of anti form of n-butane is more than its gauche form

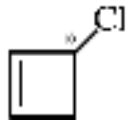
58) Based on SIR effect

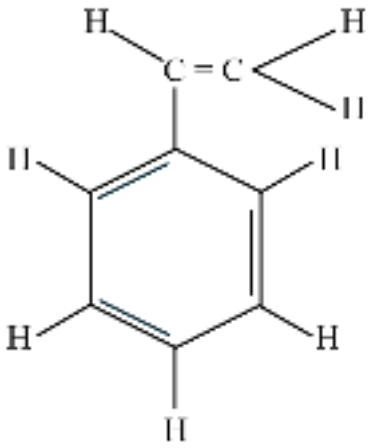
59) 
3–chloro propanal

60)

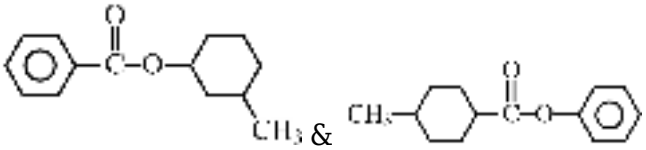
All carbon are sp^2 hybridised thus planar molecule.

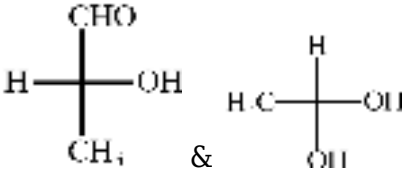
61) 
2-mercapto-5-methoxy cyclo hex-3-en-l-ol

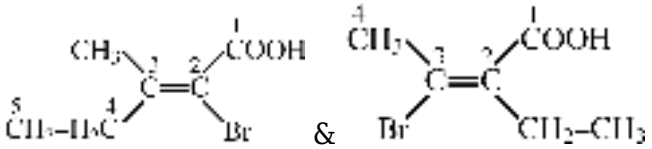
62)  is chiral

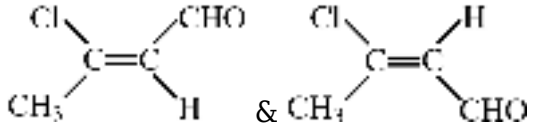
63) 
No of σ bonds = 16
No of π bonds = 4
Ratio of σ and π bonds = 16 : 4
= 4 : 1

64)

(a)  Metamer

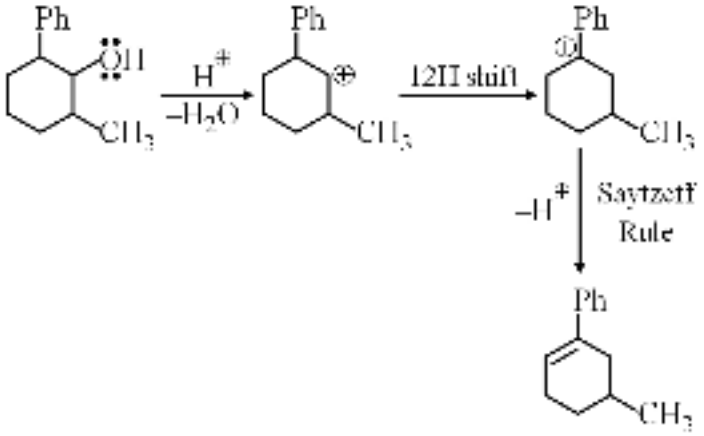
(b)  Enantiomers

(c)  Chain isomers

(d) 

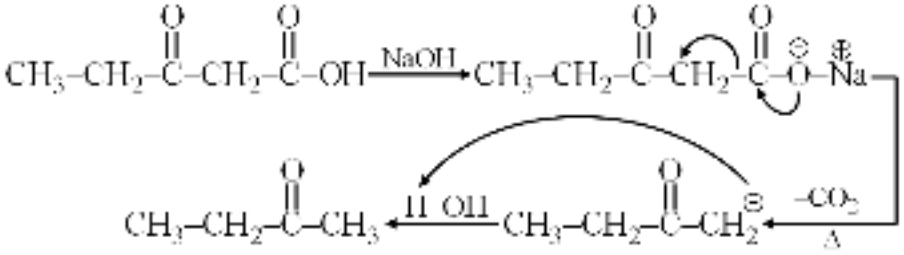
(Z) (E)

Diastereomers & Geometrical isomer

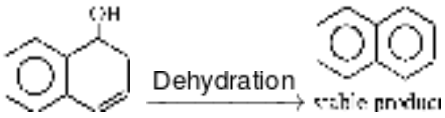
65) 

66)

Rate of decarboxylation ∝ Stability of carbanion



67) Rate of elimination ∝ Stability of carbocation ∝ Stability of alkene



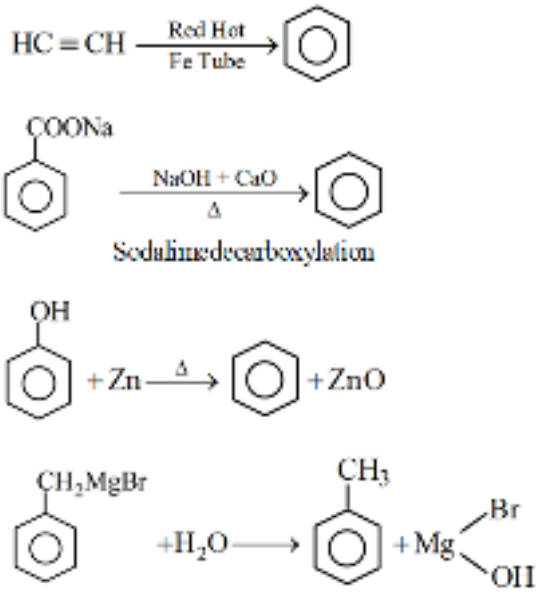
68)

$$BP \propto \text{Molecular mass} \propto \frac{1}{\text{Branching}}$$

69)

(1) E.S.R.
(2) F.R.S.R.
(3) E.S.R.
(4) E.S.R.

70)



71)

Combination of two radical is chain termination step.

72)

Rate of electrophilic addition ∝ Stability of carbocation

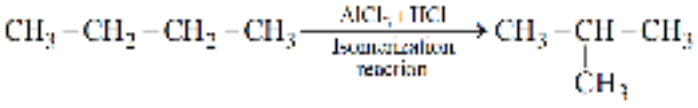
73)

$$\text{Bond dissociation energy} \propto \frac{1}{\text{Stability of free Redical}}$$

74)

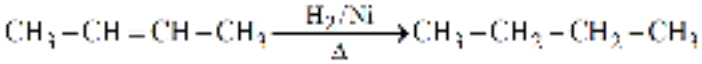
Benzene due to Aromaticity does not gives test of alkene.

75) The given reaction is characteristic of terminal alkyne.



(A)

76)

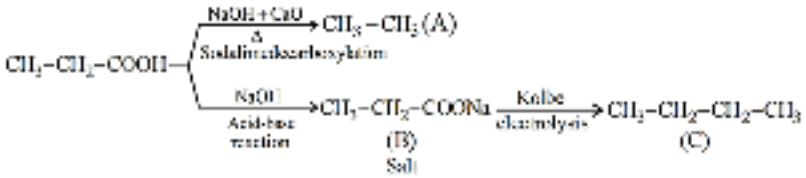


(B)

Catalytic Reduction

A and B are chain isomers

77)



78)

In free radical substitution reaction of alkane. Iodine is least reactive due to greater stability of iodine radical.

79)

Explanation:

A. Na_2O → not used.

B. Na_2SO_4 → drying agent, not adsorbent.

C. Al_2O_3 (Alumina) → widely used as an adsorbent in column chromatography.

D. NaCl → not used.

Final Answer: Option (3)

80)

C) Answer Explanation:

Steam distillation is used for compounds that are steam volatile but insoluble in water.

Such compounds co-distill with steam at lower temperatures, preventing decomposition.

Soluble or non-volatile compounds cannot be separated by this method.

D) Final Answer: Option (1) Steam volatile but insoluble in water

81)

Answer Explanation:

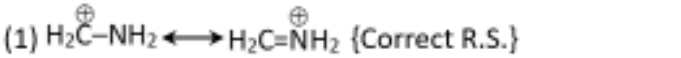
A. Nitrogen → detected as NaCN (Prussian blue test).

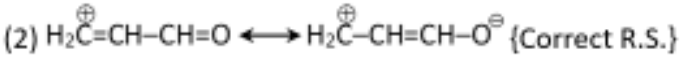
B. Sulphur → detected as Na_2S (sodium nitroprusside test).

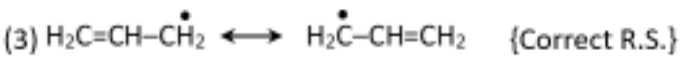
C. Halogens (Cl, Br, I) → detected as NaX (AgX ppt).

82) **Concept :**
This question is based on resonance condition.

Solution :







Final Answer :
Option (4)

83)

Electron-withdrawing groups (EWG) increase acidity; electron-donating groups (EDG) decrease it.

A. p-nitrophenol (b): Strong -R and -I effects.
B. o-nitrophenol (a): Strong -R and -I, but intramolecular H-bonding makes H^+ removal slightly harder than para.
C. m-methoxyphenol (c): -I effect of -OMe increases acidity relative to methyl.
D. m-cresol (d): +I effect of Me decreases acidity.

Final Answer: Option (3) b > a > c > d

84)

Stability follows the number of α-hydrogens (α-H):

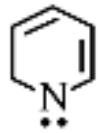
A. (a): Tetrasubstituted (10 α-H) - Most stable
B. (d): Trisubstituted (6 α-H: 3 from methyl, 1 from ring CH, 2 from ring CH_2)
C. (b): Disubstituted (4 α-H)
D. (c): Exocyclic (3 α-H) - Least stable

Final Answer: Option (4) a > d > b > c

85) The stability of a carbocation is enhanced by resonance. The carbocation with the most extensive resonance stabilization is the most stable. **Calculation**: N/A **Final Answer**: 4

86) $CH_3^{\oplus}$ due to incomplete octate and presence of vacant orbital is a electrophile.

87)

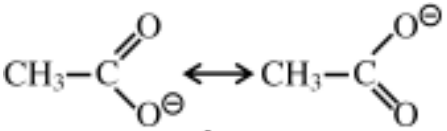
Compound		**Type**	
(A)		(Q)	Non aromatic
(B)		(P)	Aromatic
(C)		(Q)	Non aromatic

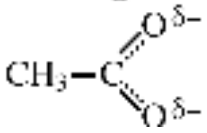
Final answer : 1

88) Due to resonance and H-effect II more stable.

89)

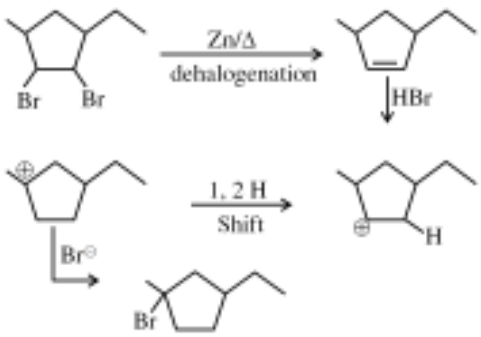
Acetate ion





≡ both are equal bond length

Acetic acid have □.p. and π resonance.

90) 

BIOLOGY

91)

NCERT-XI, Pg. # 135

92)

NCERT, Pg. # 136

93) NCERT Pg. # 137

94) NCERT, Pg # 137

95)

NCERT, Pg. # 138

96) Module No. 3 Page. No.126

97) NCERT-XI Pg. 212

98) NCERT-XI, Pg. # 213, 214, 215

99) XI NCERT Page No. # 216

100) NCERT XI Pg#219

101)

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102) NCERT Pg. # 149

103)

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104)

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105)

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106) NCERT Pg. # 154

107) NCERT Pg. # 156

108) NCERT-XI Pg. No. # 158

109) NCERT Pg. # 159

110)

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111) NCERT XI, Pg. # 233,234

112) NCERT XI Pg : 236

113)

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114) NCERT, Pg. # 230

115) NCERT Pg. # 156

116)

NCERT, Pg. # 158

117) NCERT-XI, Pg # 233

118) NCERT XI, Pg. # 233

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121) NCERT Pg. No. # 168

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125)

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130) NCERT, Pg. # 168-171

131) NCERT, Pg. # 172

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134)

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135) NCERT XI Pg. # 172, 176

136) NCERT, Pg. # 236

137) NCERT XI Pg.#236

138) NCERT Pg. # 317

139) NCERT Pg. # 234-35

140)

NCERT Pg # 231 (18)

141) NCERT Pg. # 231

142) NCERT Pg. # 232,233

143) NCERT, Pg. # 319

144) NCERT, Pg # 232, 233

145) NCERT, Pg. # 235

146) NCERT Pg. # 236

147) NCERT Pg # 236

148)

NCERT Pg. # 217

149) NCERT XI - 309, Para - 20.3

150)

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151) NCERT Pg.#219.

152) NCERT Pg. No. 219, 221

153) NCERT Pg. # 225

154) NCERT Pg. # 226

155) NCERT, Pg. # 223

156) NCERT Pg. # 221

157) NCERT Pg. # 312

158) NCERT Pg. No. # 223

159) NCERT Pg. No. # 219

160) NCERT Pg.#227

161) NCERT XI Pg # 226

162) NCERT-XI Pg#226

163)

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168) NCERT Pg # 241

169)

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170) NCERT, Pg. # 247

171) NCERT Pg. # 240

172) NCERT Pg # 239

173) NCERT Pg # 243

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177) NCERT XI Pg. 246

178) NCERT Pg. # 242

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180) NCERT, Pg. NO. 244